

Area Between Curves

ANSWER KEY



Area between two curves

- 1 Find the area of the region bounded by $y = x^2$ and $y = x + 2$.
Intersections: $x^2 = x + 2$, so $x^2 - x - 2 = 0$, $(x - 2)(x + 1) = 0$: $x = -1, 2$. Area
 $= \int_{-1}^2 [(x + 2) - x^2] dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 = \frac{9}{2}$.
- 2 The region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$ is shown. Find its area.
 $\int_0^4 \sqrt{x} dx = \left[\frac{2}{3} x^{3/2} \right]_0^4 = \frac{16}{3}$.

Setting up with respect to y, and total area

- 3 Find the area enclosed by $x = y^2$ and $x = 2y$ by integrating with respect to y .
Intersections: $y^2 = 2y$ gives $y = 0, 2$. Area $= \int_0^2 (2y - y^2) dy = \left[y^2 - \frac{y^3}{3} \right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$.
- 4 Find the total area between $f(x) = \sin x$ and the x -axis on $[0, 2\pi]$ (note the curve is below the axis on part of this interval).
By symmetry, $\int_0^\pi \sin x dx = 2$ and $\left| \int_\pi^{2\pi} \sin x dx \right| = 2$. Total area $= 2 + 2 = 4$.

More area problems

- 5 Find the area of the region bounded by $y = x^2 - 4$ and $y = 4 - x^2$.
Intersections: $x^2 - 4 = 4 - x^2$, so $2x^2 = 8$, $x = \pm 2$. Upper curve is $y = 4 - x^2$: Area
 $= \int_{-2}^2 [(4 - x^2) - (x^2 - 4)] dx = \int_{-2}^2 (8 - 2x^2) dx = \left[8x - \frac{2x^3}{3} \right]_{-2}^2 = \frac{64}{3}$.
- 6 Find the area between $y = x^3$ and $y = x$ on $[0, 1]$ (note that $y = x$ is above $y = x^3$ on this interval).
Area $= \int_0^1 (x - x^3) dx = \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$.

Areas requiring case splits

- 7 Find the area between $y = x^3$ and $y = x$ on $[-1, 1]$ (note the curves cross at $x = 0$, so the identity of the upper curve switches).
On $[-1, 0]$, $x^3 \geq x$; on $[0, 1]$, $x \geq x^3$. Area $= \int_{-1}^0 (x^3 - x) dx + \int_0^1 (x - x^3) dx$. By symmetry both integrals equal $\frac{1}{4}$ (using the earlier result), so total area $= \frac{1}{2}$.
- 8 Find the area enclosed between $y = 2x$ and $y = x^2$ on the interval where $y = 2x$ is above $y = x^2$.
Intersections: $2x = x^2$ gives $x = 0, 2$. On $(0, 2)$, $2x > x^2$: Area
 $= \int_0^2 (2x - x^2) dx = \left[x^2 - \frac{x^3}{3} \right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$.

Areas involving trigonometric and exponential curves

- 9** Find the area between $y = \cos x$ and $y = 0$ on $[0, \frac{\pi}{2}]$.

$$\text{Area} = \int_0^{\pi/2} \cos x \, dx = [\sin x]_0^{\pi/2} = 1 - 0 = 1.$$

- 10** Find the area between $y = e^x$ and $y = e^{-x}$ on $[0, 1]$ (note $e^x \geq e^{-x}$ on this interval).

$$\text{Area} = \int_0^1 (e^x - e^{-x}) \, dx = [e^x + e^{-x}]_0^1 = (e + e^{-1}) - (1 + 1) = e + \frac{1}{e} - 2.$$